The Way of the Cloud

A Study Guide for Virtual & Cloud Networking

"Before enlightenment, chop wood, carry water.
After enlightenment, configure NSGs, route packets."

January 2026

Before We Begin

> *A student once asked a master: "What is the cloud?"*
>
> *The master pointed to the sky and said nothing.*
>
> *The student looked up, saw clouds drifting, and was confused.*
>
> *The master then pointed to the student's laptop and smiled.*
>
> *"The cloud," he said, "is not up there. It is everywhere and nowhere. It is someone else's computer — and also your own."*

This guide walks the path laid out in the teaching of January 5th. It speaks of four great truths:

1. The nature of responsibility in the cloud
2. The isolation of virtual networks
3. The order of security rules
4. The two ways of balancing load

Read slowly. Pause often. Understanding comes not from speed, but from stillness.

1The First Truth: On Responsibility

The Parable of Three Houses

A traveler needed shelter for the night. They found three houses.
At the **first house**, the owner handed them an empty plot of land. "Build what you need," said the owner. "The ground is stable, but walls, roof, doors, locks — these are your concern." This is **IaaS**.
At the **second house**, the owner showed them a furnished room. "The structure stands. The bed is made. But what you do within these walls, what you bring inside, whom you let enter — this is your affair." This is **PaaS**.
At the **third house**, the owner welcomed them to a complete inn. "Eat. Sleep. Rest. We handle everything. You are but a guest." This is **SaaS**.
The traveler asked: "Which house gives me the most control?"
The owner of the first house replied: "Control and responsibility are two hands of the same body. You cannot hold one without the other."

The Essence

Infrastructure as a Service (IaaS) — You rent the foundation. You build everything else. IP addresses, routing tables, firewall rules, operating systems — all rest upon your shoulders.
Platform as a Service (PaaS) — You rent the room. The walls are built. You focus on what happens inside: your application, your data.
Software as a Service (SaaS) — You rent the experience. You simply use. The network beneath is invisible to you, maintained by unseen hands.

Contemplate: If a network misconfiguration causes an outage in IaaS, who bears responsibility?
If the same occurs in SaaS, who bears responsibility?
The answer reveals the nature of these three ways.

2The Second Truth: On Isolation

The Parable of the Walled Gardens

> In a vast kingdom called Azure, there existed many gardens. Each garden was surrounded by high walls. These were the Virtual Networks.
> A young gardener asked the kingdom's architect: "If I plant flowers in my garden, will they spread to my neighbor's garden?"
> "No," said the architect. "Each garden is complete unto itself. Your seeds cannot cross the walls. Your water cannot flow to their soil. Your garden knows nothing of theirs, and theirs knows nothing of yours."
> "But what if I wish to share my flowers?"
> "Then you must build a bridge. This bridge is called **Peering**. But know this — the bridge must be built from *both* sides. One hand alone cannot clap."
> The gardener understood: isolation is the default state. Connection requires intention.

The Three Laws of the Virtual Network

> **First Law: Isolation**
> A Virtual Network does not speak to other Virtual Networks. They exist in separate worlds, even if they share the same kingdom. Traffic between them is impossible without explicit permission.
> **Second Law: Boundaries**
> A Virtual Network belongs to one region. It cannot stretch across the earth. A network in Sweden knows nothing of a network in America, unless bridges are built.
> **Third Law: Connection Through Intention**
> VNet Peering must be configured from both sides. If VNet-A reaches toward VNet-B, but VNet-B does not reach back, the connection remains incomplete. One side shows "Connected." The other shows only "Initiated." And the packets fall into the void.

Contemplate: Two networks exist in the same region. Both use the address space 10.0.0.0/16.

A student worries: "Will they collide?"

What is the answer, knowing the nature of isolation?

3The Third Truth: On Security Rules

The Koan of the Gatekeeper

> A packet arrived at the gates of a virtual machine. The gatekeeper — the Network Security Group — consulted the book of rules.
> "Rule 100 says: Let those carrying HTTP pass."
> "Rule 200 says: Let those carrying HTTPS pass."
> "Rule 300 says: Deny all who carry HTTP or HTTPS."
> "Rule 65500 says: Deny everyone."
> The packet carried HTTP.
> The gatekeeper looked at Rule 100. The packet matched. The gatekeeper opened the gate.
> A student observed this and asked: "But Rule 300 would have denied this packet. Why did it pass?"
> The master replied: "The gatekeeper reads from the beginning, not from the end. The first match is the only match. All that follows is silence."
> **"Which rule wins?"** asked the student.
> **"The one with the smallest number,"** said the master. **"For in this world, the humble are first, and the proud are last."**

The Way of the Five Elements

Every rule in the book is judged by five elements:

Source	*Source Port*	*Destination*	*Dest. Port*	*Protocol*
Where from	Which door	Where to	Which door	What tongue

A packet must match all five elements for a rule to apply.

The Default State

> When a Network Security Group is born, it carries three truths within it:
> 1. Traffic within the Virtual Network may flow freely
> 2. Traffic going outward to the internet is permitted
> 3. **All other inbound traffic is denied**

This final truth is why a web server, freshly created, cannot be reached from the outside world. The door is closed by default. You must write a rule to open it.

A student creates two rules:

- Priority 100: Deny SSH
- Priority 200: Allow SSH

Can the student connect via SSH?

Sit with this question. The answer lies in understanding priority.

The Troubleshooting Sutra

A web server was deployed. It had a public IP address. The application ran correctly — when tested from within.

But when a user tried to reach it from the outside world, they received only silence.

"Connection timed out," spoke the browser.

The student panicked. "The server is broken!"

The master shook their head. "You can reach the server through the portal. The application runs. The problem is not within. It is at the gate."

"The NSG," whispered the student, suddenly understanding.

"Yes. The default rule denies all inbound traffic from the internet. You must add a rule — Allow, TCP, Port 80, Priority 100 — to open the way."

The student added the rule. The website appeared. The student bowed.

"Remember," said the master. "When packets disappear, first look at the gate. Use **IP Flow Verify** to see which rule speaks the final word."

4The Fourth Truth: On Balancing Load

The Parable of Two Masters

> In the kingdom of Azure, there were two masters who taught the way of distributing burdens.
> The first master worked in the **Fourth Layer**. They saw only addresses and numbers. "Tell me where you wish to go and through which port," they said to each packet. "I care not what message you carry. I will send you to a server that can bear your weight."
> This master was called **Load Balancer**.
> The second master worked in the **Seventh Layer**. They read the contents of every message. "Ah, you seek /images," they would say. "Go to the servers who specialize in images. And you, seeking /api — go to the API servers. And you with the suspicious request attempting SQL injection — you shall not pass at all."
> This master was called **Application Gateway**.
> A student asked: "Which master is greater?"
> "Neither," replied the teacher. "They serve different purposes. One sees envelopes. The other reads letters. Both are necessary. The question is not which is greater, but which is appropriate."

The Nature of Layers

> **Layer 4 — The Transport Layer**
> Here, the Load Balancer dwells. It sees:
> - IP addresses (source and destination)
> - Port numbers
> - Protocol (TCP or UDP)
>
> It does not open the envelope. It does not read the message. It simply routes based on where things are going, not what they contain.
> **Layer 7 — The Application Layer**
> Here, the Application Gateway dwells. It understands:
> - HTTP and HTTPS

- URLs and paths (/images, /api, /login)
- Headers and cookies
- The content itself

It can make intelligent decisions. It can terminate SSL. It can protect against attacks with its Web Application Firewall.

When to Choose Each Path

A database listens on port 1433 using TCP.
Which master should balance its load?
The database speaks not HTTP. It sends not URLs. It knows nothing of /paths.
Therefore, the **Load Balancer** is appropriate.

A web application has three parts:

- /images — served by image-optimized servers
- /api — served by backend API servers
- / — served by general web servers

Which master should balance its load?
The application needs routing based on URL path. Only one master reads URLs.
Therefore, the **Application Gateway** is appropriate.

The Summary of Differences

	Load Balancer	**Application Gateway**
Layer	Fourth (Transport)	Seventh (Application)
Sees	IP, Port, Protocol	HTTP content, URLs
Protocols	TCP, UDP	HTTP, HTTPS
SSL Termination	No	Yes
URL Routing	No	Yes
Web Firewall	No	Yes (WAF)
Speed	Very fast	Fast, but slower
Cost	Lower	Higher
Use for	Databases, SSH, RDP	Web apps, APIs

Note: The Basic Load Balancer completed its journey on September 30, 2025. Only the Standard Load Balancer remains. If ancient texts speak of "Basic SKU" — know that those teachings are no longer applicable.

5Final Wisdom

The Four Truths Summarized

First Truth: Responsibility flows with control.
IaaS gives you the land — you build everything, you maintain everything. As you move toward SaaS, responsibility shifts to the provider. In IaaS, you configure IP addresses, routing, and firewalls. In SaaS, you do not.
Second Truth: Virtual Networks are islands.
They do not communicate unless bridges are built. They exist in one region. Peering requires action from both sides. Isolation is the natural state.
Third Truth: The smallest number speaks first.
NSG rules are evaluated from lowest priority to highest. The first match decides the fate of the packet. Default rules deny inbound traffic. To open a door, create a rule with lower priority than the denial.
Fourth Truth: Know what you are balancing.
If it speaks HTTP, consider Application Gateway. If it speaks only TCP/UDP without application awareness, use Load Balancer. The right tool depends on the nature of the traffic.

For the Examination

When asked about responsibility: Think of who builds, who maintains, who controls.
When asked about VNet isolation: Remember that connection requires explicit configuration.
When asked about NSG priority: Lower numbers are processed first. 100 beats 200.
When asked about load balancing: Layer 4 sees envelopes. Layer 7 reads letters.
These four truths are sufficient. Master them, and the examination holds no fear.

"In the beginner's mind there are many possibilities.
In the expert's mind there are few."

— Shunryu Suzuki

6Quick Reference (The Practical Path)

For those who seek quick answers rather than contemplation:

Cloud Service Models

IaaS	Most customer responsibility. You manage OS, apps, network config.
PaaS	Shared responsibility. Provider manages infrastructure.
SaaS	Least customer responsibility. Provider manages everything.

Azure VNet

- Isolated by default — VNets cannot communicate automatically
- Regional — one VNet exists in one Azure region only
- VNet Peering requires configuration on **both** sides
- Status must show "Connected" on both sides to work

Network Security Groups

- Priority range: 100–4096 (lower = higher priority)
- **Lowest priority number is evaluated first**
- First matching rule wins; remaining rules ignored
- Default: Deny all inbound from internet
- To allow web traffic: Add rule, Allow, TCP, Port 80/443, Priority 100
- Troubleshoot with: IP Flow Verify (Network Watcher)

Load Balancer vs Application Gateway

- **Load Balancer**: Layer 4, TCP/UDP, fast, cheap, no URL routing
- **Application Gateway**: Layer 7, HTTP/HTTPS, URL routing, WAF, SSL termination

- Use Load Balancer for: databases, SSH, RDP, non-HTTP
- Use Application Gateway for: web apps, APIs, microservices